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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 11.1

$$f(x) = \frac{2x}{x^2 + 1}$$

domein f : $\mathbb{R}$

nulpunten $f(x) = 0 \Leftrightarrow x = 0$

$$\begin{array}{c|ccc} x & & 0 & \\ \hline f(x) & - & 0 & + \end{array}$$

Geen VA

$$\lim_{x \rightarrow \infty} \frac{2x}{x^2} = \lim_{\infty} \frac{2}{x} = 0$$

$$\begin{array}{c|ccc} x & & 0 & \\ \hline f(x) - A & - & | & + \end{array}$$

Er is een HA, dus er is geen SA.

$$f'(x) = \frac{(x^2 + 1) \cdot 2 - 2x \cdot 2x}{(x^2 + 1)^2} = \frac{-2x^2 + 2}{(x^2 + 1)^2}$$

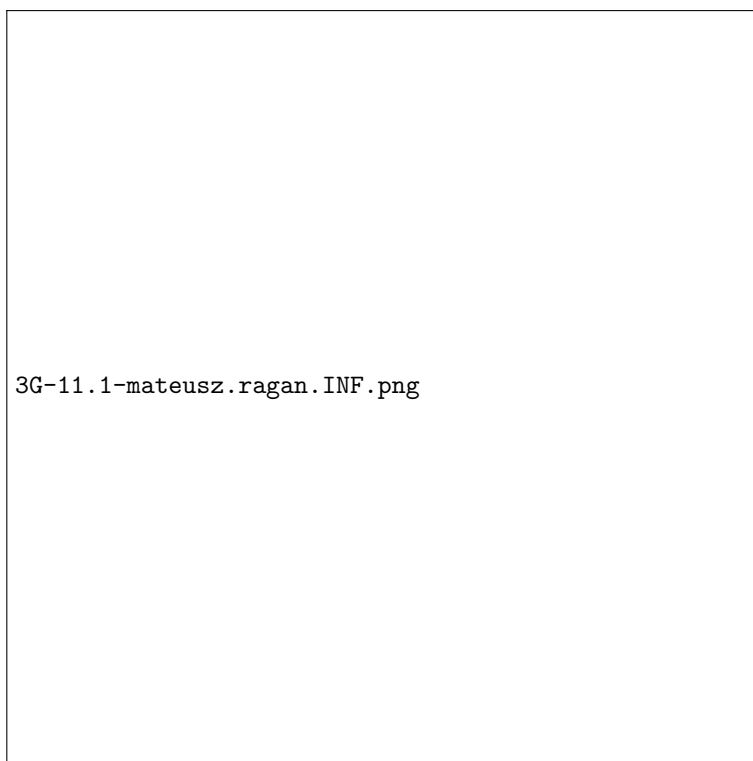
$$\begin{array}{c|ccccc} x & & -1 & & 1 & \\ \hline f'(x) & - & 0 & + & 0 & - \\ f(x) & \searrow & \min(-1) & \nearrow & \max(1) & \searrow \end{array}$$

$$f''(x) = \frac{(x^2 + 1)^2 \cdot -4x - ((-2x^2 + 2)(4x^3 + 4x))}{(x^2 + 1)^4} = \frac{4x^5 - 8x^3 - 12x}{(x^2 + 1)^4}$$

$$= \frac{4x(x^4 - 2x^2 - 3)}{(x^2 + 1)^4} = \frac{4x(x^2 + 1)(x^2 - 3)}{(x^2 + 1)^4} = \frac{4x(x^2 - 3)}{(x^2 + 1)^3}$$

$$\begin{array}{c|ccccc} x & & -\sqrt{3} & & 0 & & \sqrt{3} & \\ \hline f''(x) & - & 0 & + & 0 & - & 0 & + \\ f(x) & \frown & \text{bgpt}\left(\frac{-\sqrt{3}}{2}\right) & \smile & \text{bgpt}(0) & \frown & \text{bgpt}\left(\frac{\sqrt{3}}{2}\right) & \smile \end{array}$$

References



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